Integrated Medical Information System (EMR)

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Problem and Industry Context

The 3 Core Problems

Fragmented Medical Information - Patient records, lab results, and imaging data may be spread across different hospital systems.

Inefficient Clinical Workflows - Repeated data and multiple interfaces can decrease staff productivity and attention towards patients

Data Security - Managing sensitive patient data requires strong security and KVKK compliance

Türkiye's existing e-Nabız infrastructure provides a foundation for a potential better hospital-level integration system. Our proposed EMR focuses on integrating hospital-level systems and streamlining clinical workflows

Sources: World Health Organization; Olakotan et al.

Proposed Solution

Unified EMR System

One secure information system for hospital clinical information

Connected Data

e-Nabiz, LIS, PACS, electronic patient records

Clinical Access

Web/mobile apps, patient portal, offline syncing

Secure Records

Role-based access, 2FA, audit logs

Implementation 16 Months	Medical Tablets 4000 (estimated)	Training Sessions 200 (planned)

Sources: Olakotan et al.; U.S. Department of Health and Human Services

Why we selected the EMR system

We Had Three Options

Integrated Medical Information System	Pharmaceutical Cold-Chain Tracker	Construction RFID Tool Tracking
A secure, unified hospital information system designed for Istanbul City Hospital. Provides 24/7 access to full electronic health records in <=3 clicks (design target)	An IoT-based monitoring system designed for pharmaceutical distributors and healthcare institutions in Istanbul.	A B2B asset-management platform for construction contractors in Istanbul.
Score: 7.70 / 10 - Selected	7.25 / 10	6.60 / 10

The EMR project achieved the highest weighted score (7.70/10), outperforming the pharmaceutical cold-chain tracker (7.25/10) and RFID tool tracking system (6.60/10). Based on our selection criteria, we chose EMR for further development.

Business Model Canvas

Key Partners

- Ministry of Health (Republic of Türkiye)
- Hospital IT departments
- Hardware and Software Vendors
- Security and integration partners
- Audit / certification bodies

Key Activities

- EMR Software Development
- Integration (e-Nabiz, PACS)
- Cybersecurity testing
- National-system integration

Key Resources

- Development and IT teams
- Secure Infrastructure
- 4000 Medical Tablets
- KVKK-compliant security systems

Value Proposition

- Central, secure EMR
- Full record in ≤ 3 clicks (design target)
- Patient Portal (e-Devlet)
- Offline mobile mode
- Unified clinical workflows
- Secure patient records
- Reduces duplicate data entry

Customer Relationships

- Long-term SLA support
- Technical support and maintenance

Channels

- Web/mobile apps
- Patient Portal
- On-site deployment

Customer Segments

- Buyer: Hospital / authorized contracting body
- Users: Medical & admin staff
- Beneficiaries: Patients
- Future market: Other public hospitals

Cost Structure

- Identified direct costs: ~₺174M.
- PERT expected costs: ~₺179.2M.
- Annual OPEX: ~₺65M (preliminary).
- Full CAPEX / TCO: pending validation.

Revenue Streams

- Proposed contract: ~₺350M net.
- Milestone-based payments.
- Recurring maintenance / support fees.
- Potential public / R&D funding (unconfirmed).

Sources: Başakşehir Çam and Sakura City Hospital

RACI (Responsible, Accountable, Consulted, Informed)

Deliverable	Team Lead	Systems Architect	Integration Lead	Compliance Officer	Financial Analyst	UI/UX Specialist	Deployment Lead
EMR System Program Governance	A / R	C	C	C	C	C	C
Stakeholder Requirements	A	C	C	I	R	C	I
Security Framework	I	C	C	A / R	I	I	I
Cloud Datacenter Setup	I	A / R	C	C	I	I	I
End-Point Fleet Procurement	I	C	I	I	A / R	I	C

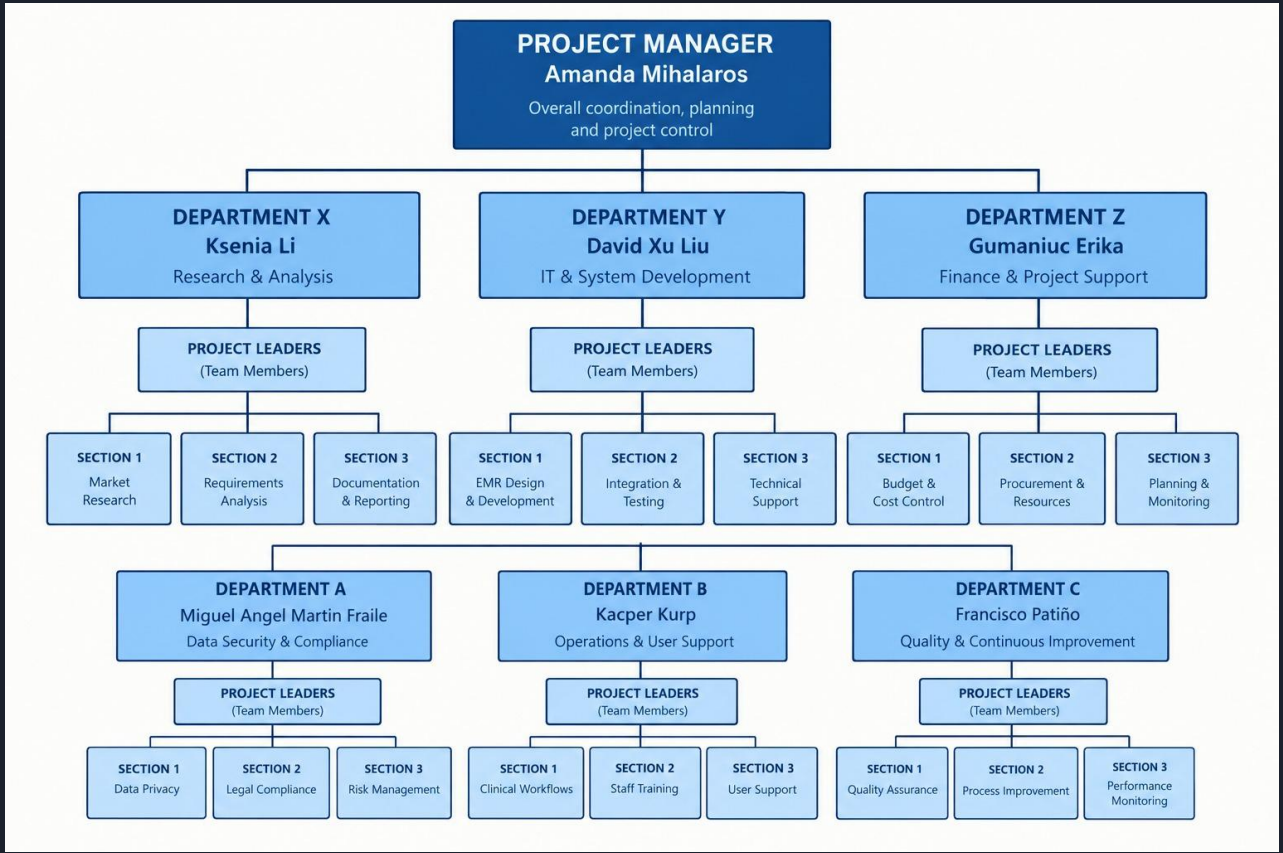
RACI (Responsible, Accountable, Consulted, Informed)

Deliverable	Team Lead	Systems Architect	Integration Lead	Compliance Officer	Financial Analyst	UI/UX Specialist	Deployment Lead
Central Database & Backend APIs	I	A / R	C	C	I	I	C
Web & Mobile	I	C	C	I	I	A / R	C
Offline Sync Engine	I	C	R	C	I	C	A
Patient Portal	I	C	C	C	I	A / R	C
National Health Integration	I	C	A / R	C	I	I	C

RACI (Responsible, Accountable, Consulted, Informed)

Deliverable	Team Lead	Systems Architect	Integration Lead	Compliance Officer	Financial Analyst	UI/UX Specialist	Deployment Lead
PACS Imaging & LIS	I	C	A / R	I	I	C	C
Legacy Data Migration	I	C	C	C	I	I	A / R
Security Audit	I	C	C	A / R	I	I	C
On-Site Training	I	I	I	I	I	A / R	C
Go-Live	A	C	C	I	I	C	R

Team structure and organisation



Roles, responsibilities and qualifications

Member	Role	Focus	FTE
Amanda Mya Mihalaros	Project Manager	Coordination, planning, control, stakeholder communication	0.50
Ksenia Li	Research & Analysis Lead	Market research, requirements, documentation	0.40
David Xu Liu	IT & System Development Lead	EMR design, integration and testing, technical support	0.50
Erika Andreea Gumaniuc	Finance & Project Support Lead	Budget and cost control, procurement, planning	0.35
Miguel Angel Martin Fraile	Data Security & Compliance Lead	GDPR, legal compliance, risk management	0.35
Kacper Piotr Kurp	Operations & User Support Lead	Clinical workflows, training, user support	0.40
Francisco Patino Guillen	Quality & Continuous Improvement Lead	Quality assurance, process improvement, KPIs	0.35

Division of work and availability

Report chapter / area	Responsible team
Introduction, user stories, requirements	Research & Analysis
Empathization, functionalities, implementation	Operations & User Support
Implementation, functionalities, integration	IT & System Development
Project idea (cost), cost estimate	Finance & Project Support
Security, GDPR, DPIA	Data Security & Compliance
Quality assurance and review	Quality & Continuous Improvement
Presentation, reporting, project control	Project Manager

Draft Work Breakdown Structure (WBS)

WBS Code	*Task Name*	*Explanations*	*Resources*	*Predecessors*
1.1	Project Initiation	Contract signing & initial tech design.	Project Manager, Architect	None
1.2.1	Cloud Datacenter	Cloud setup & deployment of 5 servers.	DevOps / Infra. Eng.	1.1
1.2.2	Fleet Procurement	Buy & deliver 4,000 medical tablets.	Procurement Lead	1.1
1.3.1	Database & API	Core EMR database development.	Lead DB Developer	1.2.1
1.3.2	Web & Mobile Apps	UI & clinician portal (≤ 3 clicks).	Senior Frontend Dev.	1.3.1
1.3.3	Offline Sync Engine	Local sync protocol for low-Wi-Fi.	Mobile App Specialist	1.3.1
1.3.4	Patient Portal	Public patient access portal.	Full-Stack Developer	1.3.1
1.4.1	National Integration	Software UAT & e-Nabız connection.	Integration Engineer	1.3.4
1.4.2	PACS & LIS Integ.	Radiology & lab results integration.	Health Systems Spec.	1.3.1
1.5.1	Data Migration	Paper digitization & DB transfer.	Data Migration Team	1.3.1
1.5.2	Security Testing	Security testing & KVKK audit.	Ext. Security Auditor	1.3.2, 1.4.1
1.6.1	On-Site Training	Staff training sessions.	Clinical Trainer Lead	1.5.2, 1.2.2
1.6.2	Go-Live & Hypercare	System launch & 24/7 helpdesk.	Support & Ops Lead	1.6.1

Action Plan

Phased Deployment Strategy & WBS-Derived Activity Schedule

Implementation Strategy

4-Phase Deployment Concept

PHASE 1 - Core System & Infrastructure (Months 1–6):

- Creating and setting up the infrastructure of medical tablets / hardware delivery
- Local cloud datacenter setup
- LAN cabling
- Firewall installation
- System integration (e-Nabız, LIS, PACS)

PHASE 2 - Software & Integrations (Months 7–9):

- Development of software, central database and user interface (UI) to meet the ≤3 clicks target per patient
- User training
- Data Migration
- Connection to national systems like e-Nabız.

PHASE 3 - Pilot Testing (Months 10–13):

- Initial testing in 3 smaller hospital departments / units
- Staff on-site training sessions
- Digitizing paper records and data migration
- Security testing

PHASE 4 - Finalization & Project Closeout (M 14–16)

- Expansion of systems to high-velocity units (ER, ICU)
- Begin the phase-out of paper systems
- 24/7 hypercare support
- Full integration of system and transition to long-term maintenance.

WBS-Derived Work Breakdown Schedule

WBS Code	Task Name	Duration	Primary Responsibility	Key Deliverables & Payment Gates
1.1	Project Initiation & Architecture	2 Mos	Project Manager / Architect	**M0 (10% Paid) upon contract signing & plan approval**
1.2.1	Local Cloud Datacenter Setup	6 Wks	DevOps / Infrastructure Eng.	**M1 (15% Paid): Cloud architecture and 5 server nodes set up**
1.2.2	End-Point Fleet Procurement	8 Wks	Hardware Procurement Lead	M4 (20% Paid): Hardware delivery of 4,000 medical tablets.
1.3.1	Central Database & Backend	12 Wks	Lead Database Developer	Build databases
1.3.2	Web & Mobile Applications	14 Wks	Senior Frontend Developer	Development of UI and clinician portal to obtain ≤ 3 clicks
1.3.3	Offline Sync Engine Module	8 Wks	Mobile App Specialist	Local sync protocol for low-Wi-Fi areas
1.3.4	Patient Portal (e-Devlet)	6 Wks	Full-Stack Developer	Public portal development
1.4.1	National Health Integration	10 Wks	Integration Engineer	**M3 (25% Paid): Software UAT integrations, connection to e-Nabız**
1.4.2	PACS Imaging & LIS Integration	8 Wks	Healthcare Systems Specialist	Integration of image viewer & lab results
1.5.1	Paper & Legacy Data Migration	8 Wks	Data Migration Team	Digitized paper documents and database transfer
1.5.2	Security & Penetration Testing	4 Wks	External Security Auditor	Security testing
1.6.1	On-Site Clinical Training	6 Wks	Clinical Trainer Lead	**M5 (10%):** training sessions completed
1.6.2	Go-Live Rollout & Hypercare	4 Wks	Support & Operations Lead	**M6 (5%):** Launch & 24/7 helpdesk

PERT Cost Estimates

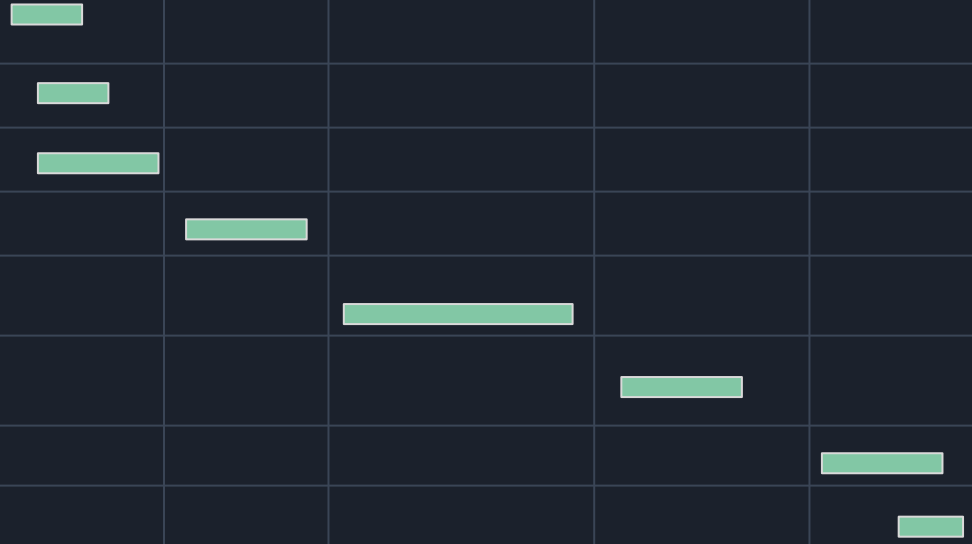
4-Phase Deployment Concept

WBS Task Name	Optimistic	Most Likely	Pessimistic	Expected PERT Cost (TRY)
1.2.1 Cloud Datacenter	10,000,000	15,000,000	25,000,000	**15,833,333 TRY**
1.2.2 Tablet	80,000,000	90,000,000	110,000,000	**91,666,667 TRY**
1.3.1 Database Development	15,000,000	20,000,000	30,000,000	20,833,333 TRY
1.3.2 Web & Mobile Apps	12,000,000	16,000,000	22,000,000	16,333,333 TRY
1.4.1 e-Nabız Integration	15,000,000	18,000,000	25,000,000	18,666,667 TRY
1.6.1 On-site Staff Training	10,000,000	15,000,000	25,000,000	15,833,333 TRY

Gantt Chart View

Project Timeline

Task	Q1 2027	Q2 2027	Q3 2027	Q4 2027	Q1 2028	Milestone
1.1 Project Initiation						M0 (10%)
1.2.1 Cloud Datacenter Setup						M1 (15%)
1.2.2 Tablet Procurement						M4 (20%)
1.3.1-4 Software						
1.4.1-2 System Integrations						M3 (25%)
1.5.1-2 Data Migration & Security						
1.6.1 On-Site Training						M5 (10%)
1.6.2 Final Implementation						M6 (5%)



Cost Baseline & Project Budget Strategy

Financial Methodology & Cost Estimation

PERT Model: Standardized weighted 3-point estimation formula used to mitigate procurement uncertainty.

Formula: Expected Cost = (Optimistic + 4 × Most Likely + Pessimistic) / 6

Governance Scope: Covers 6 critical budget work packages across hardware, infrastructure, software, and training.

Cost Baseline (PERT Calculations Table)

WBS task name	Optimistic(try)	Most likely(TRY)	Pessimistic(TRY)	Expected PERT cost(TRY)
1.2.1 Cloud Datacenter	10,000,000	15,000,000	25,000,000	**15,833,333**
1.2.2 Tablet Procurement	80,000,000	90,000,000	110,000,000	**91,666,667**
1.3.1 Database Development	15,000,000	20,000,000	30,000,000	**20,833,333**
1.3.2 Web & Mobile Apps	12,000,000	16,000,000	22,000,000	**16,333,333**
1.4.1 e-Nabız Integration	15,000,000	18,000,000	25,000,000	**18,666,667**
1.6.1 On-site Staff Training	10,000,000	15,000,000	25,000,000	**15,833,333**
Total Budget Baseline	**142,000,000**	**174,000,000**	**237,000,000**	**179,166,667 TRY**

Budget Allocation & Payment Gates

Payment Milestone Schedule

-M0 (10%): Contract signing and plan approval.

-M1 (15%): Cloud setup & 5 server nodes set up.

-M3 (25%): Software UAT integrations & e-Nabız sync.

-M4 (20%): Delivery of 4,000 medical tablets.

-M5 (10%): Clinical training sessions completed.

-M6 (5%): Official launch and 24/7 hypercare support.

Financial Feasibility Analysis

Investment Costs & Operational Expenses

- Based on our PERT cost estimation and deployment strategy, the financial commitments are:
- Capital Expenditure (CAPEX): ~179.2 Million TRY
- Annual Operational Expenditure (OPEX): ~65.0 Million TRY
- 5-Year Total Cost of Ownership (TCO): ~504.2 Million TRY

Revenue Streams & Expected Earnings

- Our financial model targets a net contract value of ~350 Million TRY over 5 years.
- Revenue Generation:

  Milestone-based contract payments (M0 to M6) for the initial build and deployment.

  Recurring annual maintenance and software licensing fees.
- 5-Year Revenue Projection:

  To achieve our net target of 350M TRY over the 504M TRY cost basis, total projected gross revenue across 5 years must reach ~854.2 Million TRY.

Profitability Measures: ROI & NPV

- Evaluating the project's 5-year financial sustainability (Discount Rate: 10%):
- Return on Investment (ROI): 195%
- The EMR system generates substantial returns relative to the initial 179.2M TRY investment, proving high financial viability over its lifecycle.
- Net Present Value (NPV): ~222.0 Million TRY
- A strong positive NPV confirms that the project's future cash flows exceed the initial investment and cost of capital, making it a highly attractive investment.

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